

# Mathematics for Data Science and Machine Learning

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*Comprehensive Topic Evaluation & Problem Bank*

## Topic 1: Foundations of Vectors & Vector Product Spaces

### 1.1 EASY

Given a data vector  $v=[3,-4,12]$  calculate its Euclidean magnitude (L2 Norm)  $\|v\|_2$ .

### 1.2 EASY

Perform vector addition for the feature inputs  $x_1=[1,5,-2]$  and  $x_2=[4,-2,6]$

### 1.3 EASY

Given a weight vector  $w=[0.4,-0.8,1.5]$ , compute the scalar product  $3w$ .

### 1.4 EASY

Determine if the target vector  $u=[1,2]$  and vector  $v=[-2,1]$  are orthogonal by computing their dot product.

### 1.5 EASY

Normalize the raw data row vector  $a=[4,3]$  into a standard unit vector  $\hat{a}$ .

### 1.6 EASY

Calculate the algebraic distance between two data points represented by vectors  $p=[1,2]$  and  $q=[4,6]$ .

### 1.7 EASY

Find the element-wise (Hadamard) product  $u \odot v$  for vectors  $u=[2,5,0]$  and  $v=[3,-1,4]$ .

**1.8 EASY**

Identify if the following structural vector entities are linearly dependent:  $v_1=[1,2]$ ,  $v_2=[2,4]$ .

**1.9 INTERMEDIATE**

Calculate the cosine similarity score between text embedding vectors  $A=[1,2,1]$  and  $B=[2,1,0]$ .

**1.10 INTERMEDIATE**

Find the projection vector of a continuous feature vector  $x=[3,5]$  onto the direction unit vector  $u=[1,0]$ .

**1.11 INTERMEDIATE**

Determine if the set of vectors  $\{[1, 0, 0], [1, 1, 0], [1, 1, 1]\}$  forms a valid basis for  $\mathbb{R}^3$  space.

**1.12 INTERMEDIATE**

Compute the angle in radians between user feature representation arrays  $u=[2,2]$  and  $v=[0,3]$ .

**1.13 INTERMEDIATE**

Verify the triangle inequality theorem  $\|x+y\| \leq \|x\| + \|y\|$  using the data coordinates  $x=[1,-3]$  and  $y=[2,4]$ .

**1.14 INTERMEDIATE**

Find a vector orthogonal to both channel feature indicators  $a=[1,1,0]$  and  $b=[0,1,1]$  using cross-product mechanics.

**1.15 INTERMEDIATE**

Given a set of observations  $S=\{[2,4],[1,2]\}$ , determine the dimensions and rank of the subspace spanned by  $S$ .

**1.16 INTERMEDIATE**

Calculate the standard Manhattan distance (L1 Norm) between error vectors  $e_1=[5,-2,3]$  and  $e_2=[1,4,0]$ .

**1.17 INTERMEDIATE**

Compute the outer product matrix expression  $u v^T$  for the single dimension column tensors  $u=[1,2,3]$  and  $v=[4,5]$ .

**1.18 DIFFICULT**

Apply the Gram-Schmidt orthogonalization pipeline process to transform the linearly independent base set  $v_1=[1,1,0]$  and  $v_2=[1,0,1]$  into an orthonormal set.

**1.19 DIFFICULT**

Prove that any set of mutually orthogonal non-zero vectors in an embedding field is always linearly independent.

**1.20 DIFFICULT**

Minimize the distance mapping expression  $\|Ax-b\|_2$  conceptually when vector  $b$  does not reside within the column space spanned by matrix operator  $A$ .

**1.21 DIFFICULT**

Derive the coordinate transformation expression for an evaluation vector  $x$  under a brand new basis matrix system  $B=\{[1,1],[0,2]\}$ .

**1.22 DIFFICULT**

Determine if the following vector space configuration satisfies Cauchy-Schwarz inequality validation for highly sparse rows:  $x=[1,0,3,0]$ ,  $y=[0,2,4,1]$ .

**1.23 DIFFICULT**

Analyze the geometric transformations when an unnormalized tensor matrix operates sequentially on an arbitrary input point vector.

**1.24 DIFFICULT**

Show that the L2 distance metric satisfies all core axioms of a true mathematical metric distance configuration space.

**1.25 DIFFICULT**

Given a multi-dimensional array vector, determine its representation under a kernelized infinite-dimensional mapping subspace.

## Topic 2: Core Matrix Operations & Linear Algebra System Layouts

### 2.1 EASY

Given matrix  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  and matrix  $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ , compute the matrix summation  $C = A + B$ .

### 2.2 EASY

For a weight matrix parameter layout  $W = \begin{bmatrix} 2 & -1 \\ 0 & 4 \end{bmatrix}$ , calculate the product under scalar coefficient scaling  $\alpha = 0.5$ .

### 2.3 EASY

Compute the standard matrix transposition  $A^T$  for the data frame layout matrix  $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ .

### 2.4 EASY

Multiply the row feature matrix  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  with the dense column vector component  $x = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$ .

### 2.5 EASY

Verify if the matrix layout  $M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  acts as the true multiplicative identity matrix for an arbitrary  $2 \times 2$  matrix structure.

### 2.6 EASY

Perform element-wise multiplication (Hadamard matrix product) for matrices  $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 5 & 0 \\ 2 & 1 \end{bmatrix}$ .

### 2.7 EASY

Determine the main diagonal trace value  $\text{Tr}(A)$  for the square parameter block layer matrix  $A = \begin{bmatrix} 10 & 3 \\ -5 & 7 \end{bmatrix}$ .

### 2.8 EASY

Identify if the following  $2 \times 2$  matrix configuration can be classified as upper triangular:  $U = \begin{bmatrix} 4 & 5 \\ 0 & 2 \end{bmatrix}$ .

### 2.9 INTERMEDIATE

Given transformation layers  $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ , calculate the exact matrix multiplication product  $C = AB$ .

### 2.10 INTERMEDIATE

Compute the explicit analytical matrix inverse  $A^{-1}$  for the given feature weights structure matrix  $A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ .

### 2.11 INTERMEDIATE

Calculate the determinant value  $\det(A)$  for the 3x3 model variance layout matrix structure:  $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 1 \\ 2 & 1 & 0 \end{bmatrix}$ .

### 2.12 INTERMEDIATE

Prove that the product sequence rule for transposes holds true by calculating  $(AB)^T$  and  $B^T A^T$  for arbitrary dimensions.

### 2.13 INTERMEDIATE

Find the rank of the following data array matrix via row reduction steps:  $X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}$ .

### 2.14 INTERMEDIATE

Determine if the following matrix is completely orthogonal by verifying if  $Q Q^T = I$  for  $Q = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ .

### 2.15 INTERMEDIATE

Solve the linear equation system represented as  $Ax=b$  using Gaussian elimination:  $A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$ ,  $b = \begin{bmatrix} 8 \\ 1 \end{bmatrix}$ .

### 2.16 INTERMEDIATE

Given a diagonal matrix  $D = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$ , find the explicit expression for computing higher matrix powers  $D^n$ s efficiently.

### 2.17 INTERMEDIATE

Construct a symmetric matrix layout by executing the operation  $S = X + X^T$  for raw feature inputs matrix  $X = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ .

### 2.18 DIFFICULT

Decompose the data layout matrix  $A = \begin{bmatrix} 4 & 3 \\ 6 & 3 \end{bmatrix}$  into its component product structures via LU decomposition mechanics.

### 2.19 DIFFICULT

Prove that if a square matrix  $A$  is invertible, its calculated determinant value  $\det(A)$  can never be equal to 0.

### 2.20 DIFFICULT

Formulate the complete analytical solution space for an underdetermined system configuration  $Ax=b$  using the pseudo-inverse formulation matrix operator.

### 2.21 DIFFICULT

Show that the trace operations map satisfies the cyclic property identity rule:  
 $\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB)$ .

### 2.22 DIFFICULT

Analyze the computational complexity variations when evaluating nested matrix multiplications of shapes  $(N,K) \times (K,M) \times (M,P)$  dynamically.

### 2.23 DIFFICULT

Prove that the non-zero eigenvalues of product matrix form  $AB$  match perfectly with the eigenvalues of product matrix form  $BA$ .

### 2.24 DIFFICULT

Derive the Sherman-Morrison rank-1 matrix inversion update equation formula used extensively in iterative optimization tasks.

### 2.25 DIFFICULT

Analyze the numeric stability profile of performing matrix inversions via Cramer's rule versus standard QR factorizations.

## Topic 3: Advanced Matrix Transformations & Linear Subspaces

### 3.1 EASY

Identify the dimensions of the product matrix if a matrix of shape  $3 \times 2$  multiplies another matrix of shape  $2 \times 5$ .

### 3.2 EASY

Compute the squared matrix block layer  $A^2$  for the small kernel weights matrix  $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ .

### 3.3 EASY

Given a row matrix vector  $r = [1, -2]$  and column matrix vector  $c = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ , find the product scalar value  $rc$ .

### 3.4 EASY

Show if the identity matrix operation preserves any vector direction state under basic linear dot products.

### 3.5 EASY

Extract the column sub-components from the training data batch frame given by matrix  $M = \begin{bmatrix} 5 & 1 \\ 3 & 2 \end{bmatrix}$ .

### 3.6 EASY

Find the negative inverse configuration scaling mapping result for matrix  $-1 * \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$ .

### 3.7 EASY

Determine the spatial dimension values for a linear system block of dimensions  $100 \times 50$  operations layer.

### 3.8 EASY

Check if matrix multiplication is commutative by comparing product layouts  $AB$  and  $BA$  for arbitrary  $2 \times 2$  blocks.

### 3.9 INTERMEDIATE

Calculate the characteristic polynomial expression matrix formulation for the transformation matrix structure  $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$ .

### 3.10 INTERMEDIATE

Extract the exact numerical Eigenvalues  $\lambda$  for the data covariance matrix template matrix  $\Sigma = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ .

### 3.11 INTERMEDIATE

For the calculated eigenvalues of matrix  $M = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ , find their corresponding normalized system Eigenvectors.

### 3.12 INTERMEDIATE

Find the exact projection plane transformation matrix layout that projects any arbitrary 3D spatial data item onto the 2D xy-plane area.

### 3.13 INTERMEDIATE

Determine the explicit fundamental Null Space configuration basis for the system matrix layer  $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$ .

### 3.14 INTERMEDIATE

Verify the Rank-Nullity Theorem balance identity rule using an operations layer matrix with rank 2 and nullity dimensions 1.

### 3.15 INTERMEDIATE

Diagonalize the square weight parameter block layer matrix  $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$  into its component product form  $P D P^{-1}$ .

### 3.16 INTERMEDIATE

Find the orthogonal complement space structure for a line vector subspace spanned by direction vector  $v = \begin{bmatrix} 1 & 2 \end{bmatrix}$ .

### 3.17 INTERMEDIATE

Compute the singular values indicators for the rectangular matrix system block structure matrix  $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ .

### 3.18 DIFFICULT



Perform Singular Value Decomposition (SVD) analysis to break down the transformation tracking matrix  $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$  into components  $U \Sigma V^T$ .

### 3.19 DIFFICULT

Prove that symmetric covariance data structures always possess completely real eigenvalues and orthogonal eigenvectors properties.

### 3.20 DIFFICULT

Compute the Moore-Penrose generalized pseudo-inverse matrix form  $A^+$  for a singular non-invertible data matrix column layout.

### 3.21 DIFFICULT

Analyze the geometric expansion indicators of the determinant operation under multi-dimensional mapping stretches.

### 3.22 DIFFICULT

Derive the mathematical proof establishing the spectral decomposition theorem structure for compact symmetric operators matrices.

### 3.23 DIFFICULT

Show how Principal Component Analysis (PCA) directly utilizes the eigenvector transformations of empirical data covariance structures.

### 3.24 DIFFICULT

Formulate the algorithmic processing mechanics of transforming feature dimensions using randomized low-rank matrix approximations.

### 3.25 DIFFICULT

Evaluate the propagation of floating-point rounding errors when executing Householder reflections during QR matrix decomposition.

## Topic 4: Multi-Dimensional Functions, Mappings & Kernel Spaces

### 4.1 EASY

Evaluate the standard sigmoid activation function mapping expression  $\sigma(z) = 1 / (1 + e^{-z})$  for the input value  $z = 0$ .

### 4.2 EASY

Compute the output value for a standard Rectified Linear Unit (ReLU) activation function  $f(x) = \max(0, x)$  for input  $x = -4.5$ .

### 4.3 EASY

Given a linear layer activation function transformation mapping model  $f(x) = 3x + 2$ , evaluate the functional value output for input scalar  $x = 4$ .

### 4.4 EASY

Identify if the function transformation mapping expression  $f(x) = x^2$  satisfies the properties required to be a linear mapping.

### 4.5 EASY

Evaluate the binary step function mapping for an activation threshold layer assuming input parameter  $x = 0.5$ .

### 4.6 EASY

Compute the coordinate output of an affine transformation function shift  $f(x) = Ax + b$  where  $A=I$ ,  $x=[1,2]$ ,  $b=[3,1]$ .

### 4.7 EASY

Find the inverse function map expression  $f^{-1}(y)$  for the invertible target linear layer function model  $y = 2x - 5$ .

### 4.8 EASY

Evaluate the hyperbolic tangent activation function map  $\tanh(z)$  at the central domain position parameter  $z = 0$ .

#### 4.9 INTERMEDIATE

Determine if the multivariate mapping layout function expression  $T(x, y) = (2x + y, x - y)$  forms a valid linear transformation structure.

#### 4.10 INTERMEDIATE

Calculate the localized function derivative rate of change for the activation function profile  $f(x) = \ln(1 + e^x)$  (Softplus activation mapping).

#### 4.11 INTERMEDIATE

Compute the multi-dimensional output array vector for a Softmax activation layer given input score values vector  $z = [1.0, 2.0]$ .

#### 4.12 INTERMEDIATE

Evaluate a radial basis function (RBF) kernel mapping expression  $K(x, c) = e^{-||x-c||^2}$  given point vector  $x = [1, 2]$  and center vector  $c = [1, 1]$ .

#### 4.13 INTERMEDIATE

Construct the explicit transformation matrix coordinate layer tracking form representing a 2D spatial rotation mapping by angle  $\theta = \pi/2$ .

#### 4.14 INTERMEDIATE

Find the structural composition function product mapping result  $f(g(x))$  for functions  $f(u) = u^2$  and function model  $g(x) = 2x + 1$ .

#### 4.15 INTERMEDIATE

Analyze the geometric stretching effects when an affine function map translates a normalized input sphere cluster along a constant bias direction vector.

#### 4.16 INTERMEDIATE

Determine the domain constraints and parameter ranges for a log-odds logit transform mapping function expression  $\log(p / (1-p))$ .

#### 4.17 INTERMEDIATE

Verify if a homogeneous transformation matrix layout can encapsulate both spatial rotations and translations within an integrated vector product system.

**4.18 DIFFICULT**

Prove that a polynomial kernel function mapping structured as expression  $K(x, y) = (x^T y + c)^d$  always satisfies Mercer's theorem criteria for valid reproducing kernel Hilbert spaces (RKHS).

**4.19 DIFFICULT**

Derive the multi-dimensional gradient matrix mapping vector expression for the generalized categorical cross-entropy loss function coupled with a Softmax layer.

**4.20 DIFFICULT**

Analyze the mathematical vanishing gradient problem by computing higher-order chain rules on nested sigmoid composition functions.

**4.21 DIFFICULT**

Formulate the structural transformation rules for maps operating over infinite-dimensional continuous functional spaces.

**4.22 DIFFICULT**

Prove that any arbitrary linear mapping between finite-dimensional spaces can be represented uniquely as a coordinate tensor matrix product operation.

**4.23 DIFFICULT**

Evaluate the geometric distortion indicators of non-linear structural manifolds under localized local linear transformations approximations.

**4.24 DIFFICULT**

Derive the mathematical boundaries of the optimization space when mapping feature dimensions into high-dimensional space via an explicit custom mapping function string.

**4.25 DIFFICULT**

Prove the mathematical convergence guarantees of training parameterized functional mappings under Lipschitz continuity constraint boundaries.